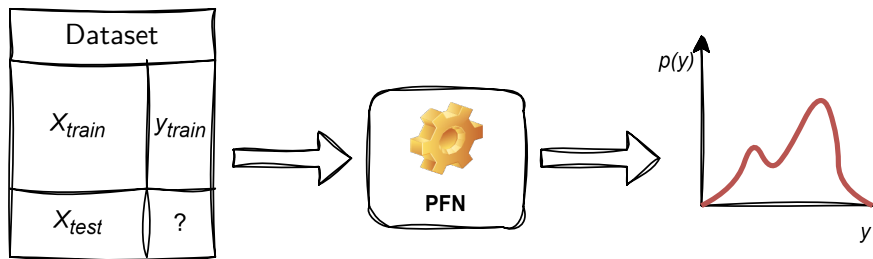


Incorporating graph knowledge into TabPFN

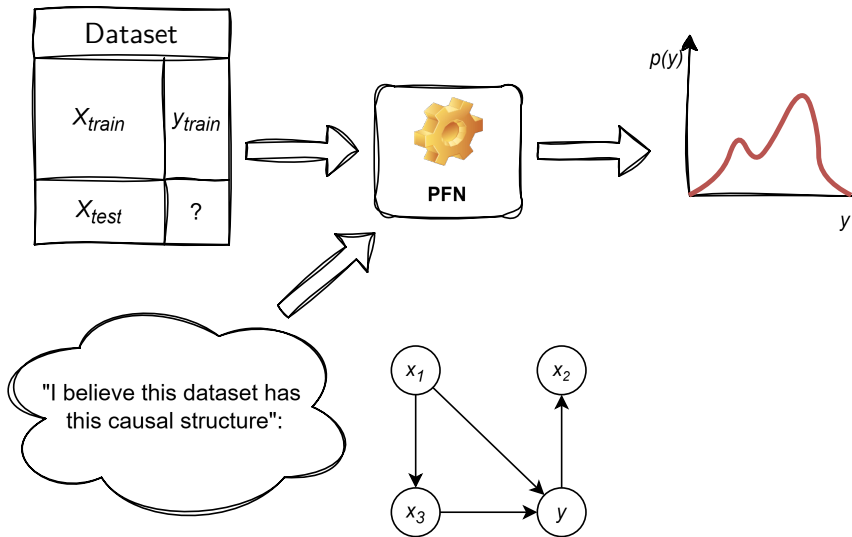
Ole Ossen

February 26, 2026

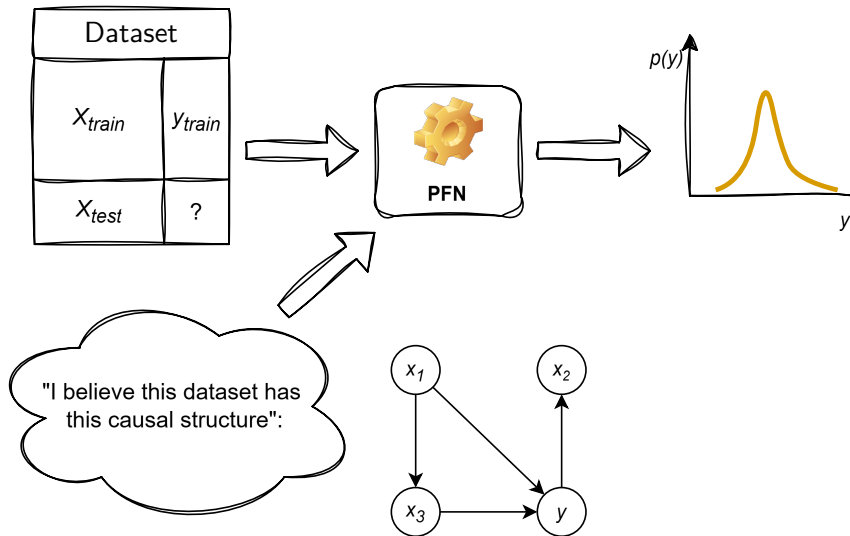
Idea



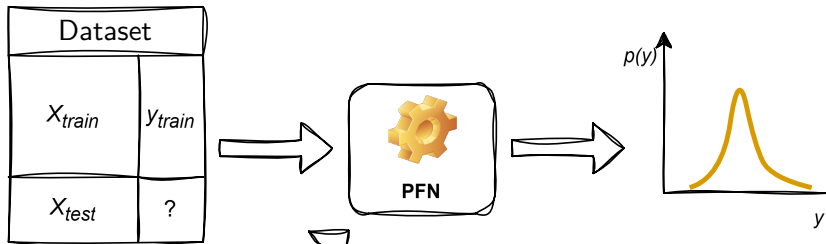
Idea



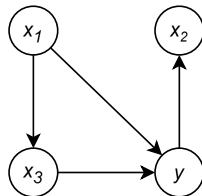
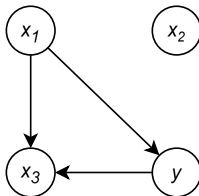
Idea



Idea



"I believe this dataset has this causal structure. ...or maybe this one?"



Graph information

The “graph information” is a probabilistic adjacency matrix

$$\gamma \in [0, 1]^{(F+1) \times (F+1)}, \quad \gamma + \gamma^t \in [0, 1]^{(F+1) \times (F+1)}.$$

Given nodes i, j , there is

- an edge $i \rightarrow j$ with probability γ_{ij}
- an edge $j \rightarrow i$ with probability γ_{ji}
- no edge between i and j with probability $1 - \gamma_{ij} - \gamma_{ji}$

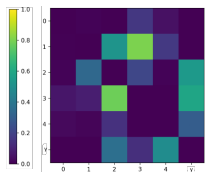
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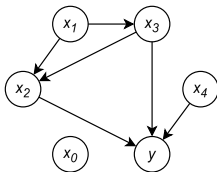
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graph info γ

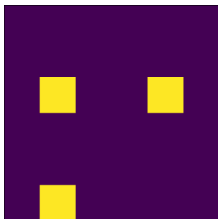


graph



Graph information

Graph information can be at different levels of uncertainty:



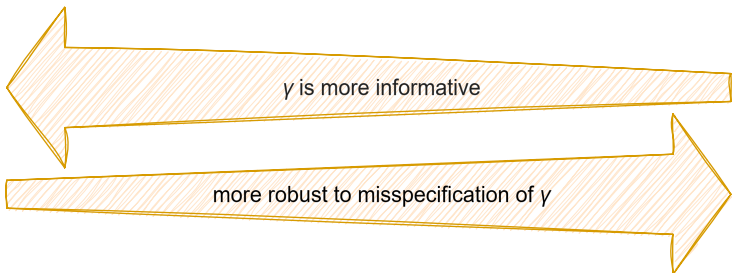
"binary"



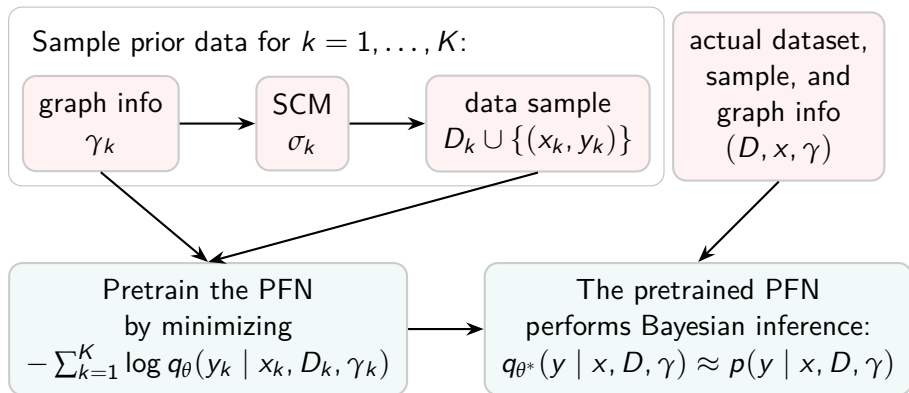
"beta"



"uncertain"



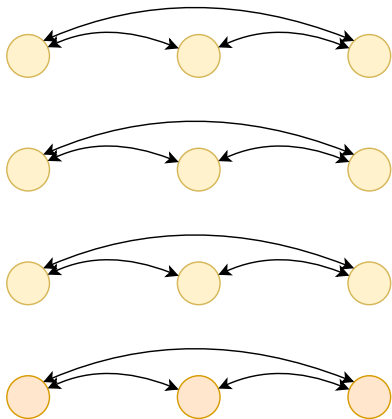
Foundation model overview



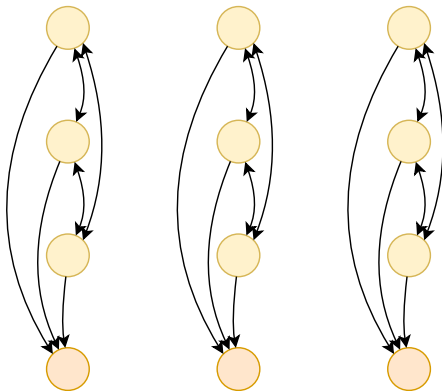
(based on an analogous illustration in Müller et al., 2024)

Architecture

Feature attention

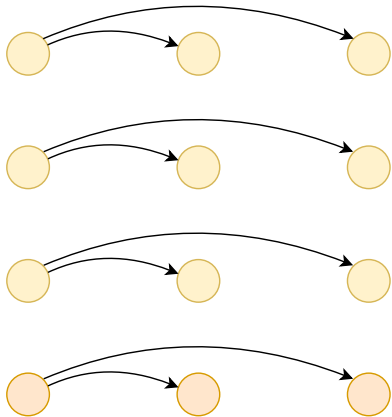


Sample attention

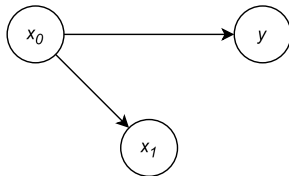


Architecture

Feature attention



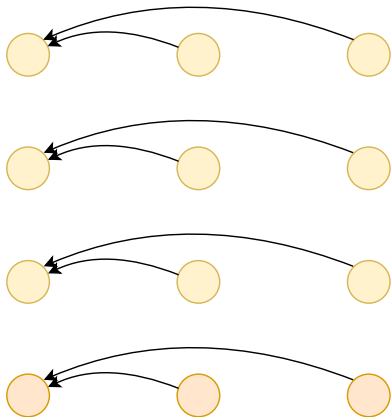
Idea: restrict feature attention based on adjacency in causal graph



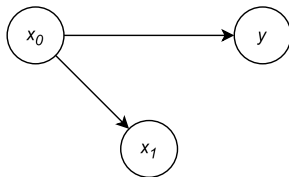
For some heads, attention is only possible to **children** in the causal graph;
for other heads, attention is only possible to **parents** in the causal graph

Architecture

Feature attention



Idea: restrict feature attention based on adjacency in causal graph



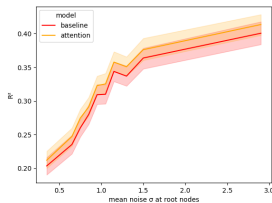
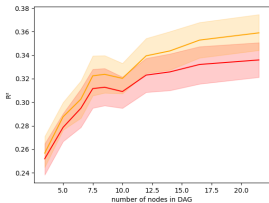
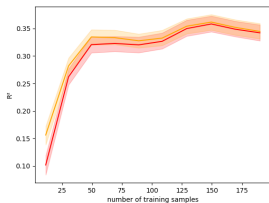
For some heads, attention is only possible to **children** in the causal graph;
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Architecture

- How does this work if graph information is not binary?
 - ▶ Add penalty to attention weight based on the entries $\gamma_{ij} \in [0, 1]$
 - ▶ I use $\log \gamma_{ij}$, which gives binary attention masking in case $\gamma_{ij} \in \{0, 1\}$
- Every node is allowed to attend to itself
- Optionally have some heads that are allowed to attend to everything
- Alternative to restricted attention: Find graph embedding $\varphi(\gamma) \in \mathbb{R}^{F+1}$ and add to cell embeddings
 - ▶ I experimented with a graph convolutional network (GCN); it worked okay

Experiments

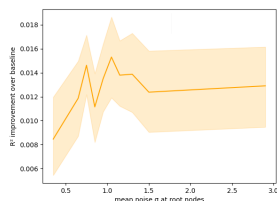
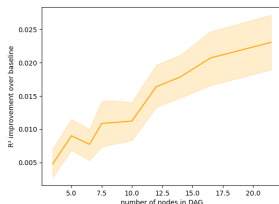
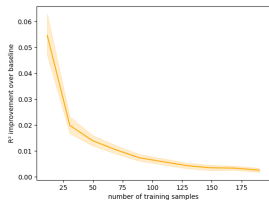
Performance of a model with “beta” type graph information against baseline, as a function of dataset characteristics:



The task becomes easier with more context, more nodes in DAG, more varied input (showing 95% bootstrapped confidence intervals).

Experiments

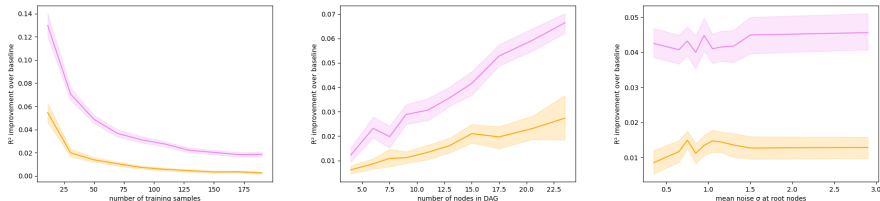
Improvement when incorporating “beta” graph information against non-graph-conditioned baseline:



The improvement is larger with small context and many nodes.

Experiments

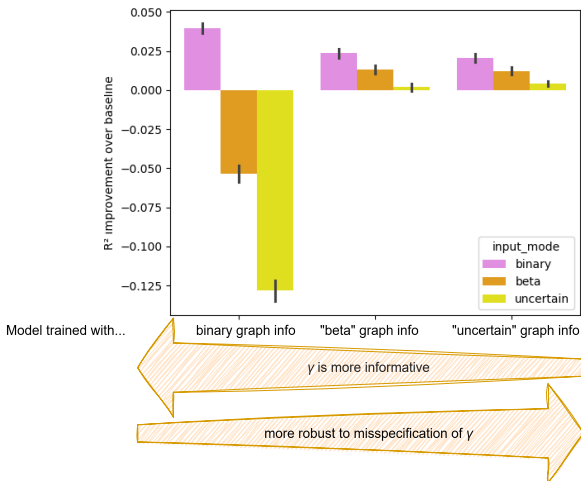
Improvement when incorporating “beta” graph information compared with improvement when incorporating binary graph information:



Gains are larger with binary graph information. Is binary graph information just better?

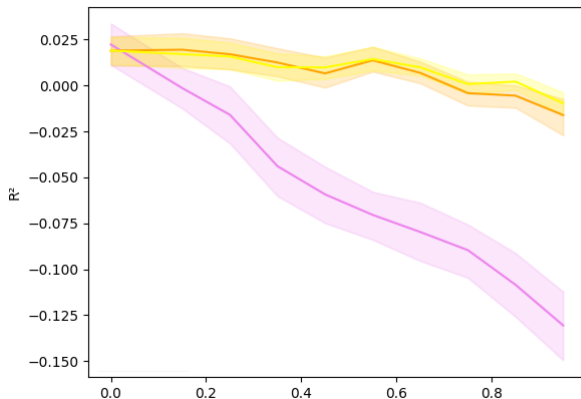
Varying uncertainty level of graph information

Not really! Recall the tradeoff between performance and robustness when choosing graph information uncertainty:



Misspecifying graph information

Performance degradation as a function of the fraction of edges replaced with “I don’t know”, for models trained with “uncertain”, “beta”, and binary graph information:



Training against soft labels

Usually we train PFNs to minimize negative log-likelihood:

$$\text{NLL}_\theta = \mathbb{E}_{D \cup \{x, y\}} \left[-\log q_\theta(y \mid x, D) \right]$$

Fact (Müller et al., 2024)

Up to an additive constant, NLL_θ is the same as

$$\mathbb{E}_{D, x} \left[\text{KL}(p(\cdot \mid x, D), q_\theta(\cdot \mid x, D)) \right]. \quad (1)$$

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If we know the distributions $p(\cdot \mid x, \sigma)$, we can instead minimize

$$\text{CEL}_\theta = \mathbb{E}_{\sigma, D, x} \left[H(p(\cdot \mid \sigma, x), q_\theta(\cdot \mid x, D)) \right].$$

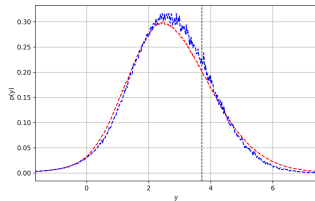
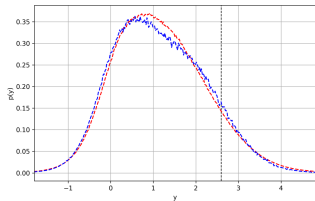
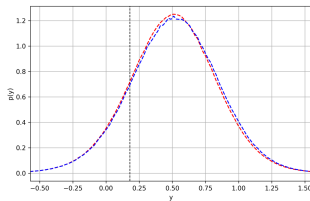
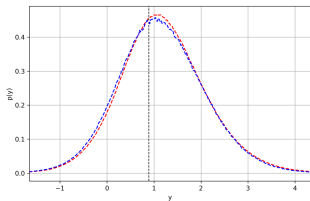
Fact

Up to an additive constant, CEL_θ is also the same as (1)!

Training against soft labels

With an additive noise model (ANN) prior, the $p(\cdot \mid x, \sigma)$ are available.

The difference between a model trained **with CEL_θ** and a model trained **with NLL_θ** is visible to the naked eye:



Training against soft labels

...but not to all metrics:

- R^2 winrate: $51.9\% \pm 1\%$
- MSE winrate: $53.08\% \pm 1\%$
- NLL winrate: $54.11\% \pm 1\%$
- CEL winrate: $57.11\% \pm 1\%$
- Winrate with respect to smoothness $\int_{\mathbb{R}} p'(y)^2 dy$: $92.78\% \pm 0.5\%$

Training using CEL_{θ} is much more data efficient, so possibly big gains in pretraining speed:

- ≈ 3 times speedup in TabPFN regressor speedrun challenge
- But interaction with number of query samples and batch size is underexplored

Discussion

- Graph conditioning works and is useful
 - ▶ ...at least for small sample size
- It's not just a “trick”, the model predicts what is right according to the Bayesian formalism
- Another use case: interventional setting
 - ▶ New preprint on this just out (Reuter et al.)
- Potentially bigger gains in OOD setting
 - ▶ But what exactly does this even mean?
 - ▶ How should train and test distribution differ? What graph info(s) do we provide?
- Training against soft labels is interesting, but it's not obvious just how much or where it helps